Design and Analysis of a Standalone Solar PV-Fed IFOC Induction Motor Drive for Water Pumping

1. Introduction

1.1 Background and Motivation

Water pumping is an essential requirement in agricultural irrigation, livestock management, and domestic water supply. In many rural and remote regions, however, access to a reliable electrical grid may be limited or completely unavailable. Conventional water-pumping systems in such locations often depend on diesel-powered generators or grid-connected electric pumps. Diesel-based systems require continuous fuel supply and regular maintenance, while extending the electrical grid to isolated locations can involve significant infrastructure and installation costs.

Solar photovoltaic (PV) energy provides an attractive alternative for such applications. A PV array converts incident solar radiation directly into electrical energy and can therefore supply a water-pumping system without requiring a continuous external fuel source. The availability of solar energy during daytime hours also corresponds well with many irrigation and water-storage applications, where water can be pumped when solar energy is available and stored for later use.

However, a solar PV source differs fundamentally from a conventional regulated electrical supply. The voltage, current, and maximum available power of a PV array depend on environmental conditions, particularly solar irradiance and cell temperature. Furthermore, the PV array exhibits a nonlinear current-voltage characteristic and can deliver maximum power only within a particular operating region. Consequently, directly connecting a motor-pump system to a PV array does not provide effective utilization of the available solar energy or controlled operation of the motor.

A power-electronic conversion and control system is therefore required between the PV source and the motor. Such a system must extract the available energy from the PV array, establish a suitable DC-link voltage for the motor drive, and control the motor according to the required pumping operating point. These requirements make a standalone solar PV water-pumping system a coupled energy-conversion and control problem involving the PV source, power converters, motor drive, and mechanical load.

1.2 Standalone Solar PV-Fed Water Pumping Systems

A standalone solar PV-fed water-pumping system operates independently of the electrical grid, with the PV array serving as the primary source of electrical energy. Since the voltage produced by the PV array may not be directly suitable for supplying the motor-drive inverter, a DC-DC converter is generally introduced between the PV array and the DC link. The DC-link voltage is then converted into a variable-voltage, variable-frequency three-phase supply through a voltage-source inverter (VSI) to drive the electric motor.

The general power-conversion structure considered in this project is

PV Array → Boost Converter → DC Link → Three-Phase VSI → Induction Motor → Centrifugal Pump

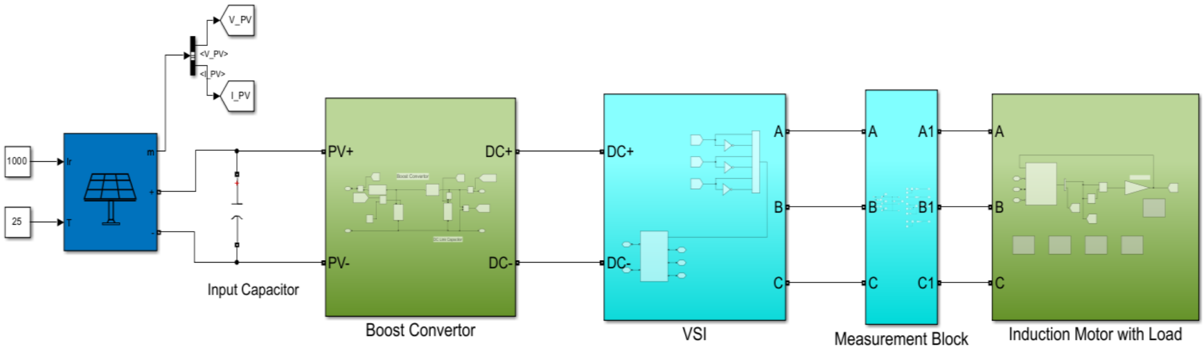

Figure 1.1: Overall Architecture of the Standalone Solar PV-Fed Induction Motor Water-Pumping System

The boost converter provides an interface between the PV array and the DC link and allows the PV operating point to be influenced through its duty ratio. The DC-link capacitor acts as an intermediate energy-storage element and reduces the instantaneous power imbalance between the source and the inverter. The three-phase VSI subsequently supplies the induction motor according to the voltage commands generated by the motor-control system.

The use of an induction motor provides a robust and widely established solution for pumping applications. However, obtaining controlled dynamic operation requires suitable regulation of the motor torque, flux, and speed. In the present system, this is achieved using Indirect Field-Oriented Control (IFOC), which enables the flux-producing and torque-producing components of the stator current to be controlled approximately independently.

The integration of these subsystems introduces an important control challenge. From the PV-source perspective, operation near the maximum-power point is desirable in order to utilize the available solar energy effectively. From the motor-drive perspective, however, the DC-link voltage must remain sufficiently high for the inverter to generate the voltage demanded by the motor controller. Therefore, the desired PV operating point cannot be considered independently of the instantaneous requirements of the motor drive.

This interaction between maximum-power extraction and the minimum DC-link voltage required for motor operation forms one of the central considerations in the development of the proposed system. The control challenges arising from these competing operating requirements are discussed in the following section.

1.3 Control Challenges in Solar PV-Fed Motor Drives

The integration of a photovoltaic source with a controlled motor drive introduces a fundamental coordination problem between the source and the load. Unlike an ideal DC voltage source, a PV array has a nonlinear current-voltage characteristic and a finite amount of power available under a given irradiance and temperature condition. The operating voltage of the PV array therefore directly influences the amount of power that can be extracted from it.

From the source-side perspective, it is desirable to operate the PV array near its maximum-power point (MPP), where the maximum available power can be extracted under the prevailing environmental conditions. In conventional PV systems, a maximum power point tracking (MPPT) controller is commonly employed to continuously adjust the converter operating point toward this condition.

The motor-drive system, however, introduces an additional requirement. The voltage-source inverter can generate the commanded three-phase motor voltages only when the DC-link voltage is sufficiently high. As the motor operating point changes, the voltage demanded by the vector-control system also changes. Consequently, the minimum DC-link voltage required for satisfactory motor operation is not necessarily constant but depends on the instantaneous voltage requirement of the drive.

A control strategy based only on maintaining a fixed DC-link voltage may ensure sufficient inverter voltage capability but does not directly account for the operating point at which the maximum available PV power can be extracted. Conversely, a control strategy concerned only with maximum-power operation may not explicitly ensure that the DC-link voltage remains sufficient to satisfy the voltage requirement of the motor drive.

The source and load sides therefore cannot be treated as completely independent systems. The boost-converter operating point influences both the PV-side power extraction and the DC-link voltage available to the inverter. An effective control strategy must consequently coordinate these two requirements while respecting the physical power capability of the PV source.

This motivates the development of a supervisory operating strategy in which the minimum DC-link voltage required by the motor drive is determined from the inverter voltage commands and considered together with the maximum-power operating objective of the PV array.

1.4 Proposed System and Project Objectives

The objective of this project is to design, develop, and evaluate a standalone solar PV-fed induction motor drive for a centrifugal water-pumping application in MATLAB/Simulink. The work focuses on the coordinated operation of the PV source, power-electronic conversion stages, and vector-controlled induction motor drive under varying operating conditions.

The principal objectives of the project are:

- To design and size the PV array, boost converter, input capacitor, and DC-link capacitor according to the system power and voltage requirements.
- To develop an Indirect Field-Oriented Control scheme for speed control of the induction motor driving a centrifugal pump load.
- To determine the minimum DC-link voltage required by the motor drive from the instantaneous inverter voltage commands.
- To develop a supervisory control strategy that coordinates the minimum DC-link voltage requirement with the maximum-power operating objective of the PV array.
- To integrate the complete PV-fed motor-drive system and evaluate its performance under variations in irradiance, temperature, and speed reference.
- To investigate the sustainable operating limits of the standalone system and analyse the behaviour of the PV source and DC link when the demanded load power exceeds the available source capability.

The scope of the present work is limited to simulation-based design and validation in MATLAB/Simulink. Rotor speed is directly available to the IFOC system, and sensorless speed estimation is therefore outside the scope of the present project.

2. System Architecture and Component Design

2.1 System Specifications

The design of the standalone solar PV-fed water-pumping system is primarily determined by the power and operating requirements of the induction motor. The motor selected for this project is a three-phase squirrel-cage induction motor with a rated mechanical output power of 4 kW (5.4 HP), a rated line-to-line voltage of 400 V, a rated frequency of 50 Hz, and a rated speed of 1430 rpm. The motor drives a centrifugal pump, whose load characteristics are modelled based on the rated operating point in a subsequent section.

The induction motor is supplied through a three-phase voltage-source inverter connected to the DC link. To provide the voltage required by the inverter for satisfactory operation of the 400 V motor, a nominal DC-link voltage of 670 V is considered for the power-stage design. This voltage serves as the nominal operating point for the initial design and analysis of the boost converter and DC-link components.

Since the voltage available from the PV array is lower than the required DC-link voltage, a boost converter is employed between the PV source and the DC link. A switching frequency of 20 kHz is selected for the boost converter and is used in the subsequent design of its passive components.

These initial specifications establish the voltage and power requirements of the complete system. Based on these requirements, the PV array configuration, boost-converter components, DC-link capacitor, inverter, and centrifugal pump load are designed and selected in the following sections.

2.2 PV Array Sizing and Configuration

The PV array serves as the sole electrical energy source for the standalone water-pumping system. Its series and parallel configuration must therefore provide a suitable input-voltage range for the boost converter while also providing sufficient power to supply the 4 kW motor-drive system.

The nominal DC-link design voltage is 670 V. Supplying the boost converter from a very low PV voltage would require operation at a high duty ratio, resulting in a large voltage-conversion ratio and less desirable converter operating conditions. Therefore, a PV array operating voltage of approximately 350 V was selected to provide a moderate voltage step-up between the PV array and the DC link.

The selected PV module has a maximum-power-point voltage of

$$V_{mpp} = 29 \text{ V}$$

To obtain a PV operating voltage close to the desired value of 350 V, 12 modules are connected in series. The resulting array voltage at the maximum-power point is

$$V_{PV,mpp} = 12 \times 29 = 348 \text{ V}$$

The current of a single PV module at the maximum-power point is 7.35 A. Since series connection increases the array voltage while maintaining the current of a single module, one series string provides approximately 7.35 A. To increase the available power capability of the array, two identical strings are connected in parallel, giving

$$I_{PV,mpp} = 2 \times 7.35 = 14.7 \text{ A}$$

The resulting maximum power available from the complete array under standard test conditions is therefore approximately

$$P_{PV,mpp} = 348 \times 14.7 \approx 5.12 \text{ kW}$$

The final PV array therefore consists of 12 modules connected in series per string and two such strings connected in parallel, giving a total of 24 modules. The resulting rated maximum-power capability provides sufficient capacity for the 4 kW motor-drive system while maintaining a PV-side voltage suitable for the required boost-converter voltage conversion.

The use of maximum-power-point quantities in the above sizing represents the rated useful power capability of the PV array under standard environmental conditions. It does not imply that the array is constrained to operate continuously at the maximum-power point. In the final integrated system, the PV operating point is determined by the supervisory control strategy according to both the available PV power and the DC-link voltage requirement of the motor drive.

2.3 PV-Side Input Capacitor Design

The switching operation of the boost converter results in a pulsating current demand at its input. Without an input-side capacitor, a significant portion of this high-frequency current ripple would be imposed directly on the PV array, resulting in increased ripple in the PV terminal voltage. An input capacitor is therefore connected across the PV array terminals to provide a local path for the switching-frequency current components and maintain a comparatively smooth PV voltage.

The required input capacitance is estimated using

$$C_{in} = \frac{I_{in} D}{f_s \Delta V_{PV}}$$

where I_{in} is the maximum considered input current, D is the nominal boost-converter duty ratio, f_s is the switching frequency, and ΔV_{PV} is the allowable ripple in the PV voltage.

Since the input capacitor must accommodate operation at the maximum expected PV current, the array maximum-power-point current of 14.7 A is considered for the capacitor design.

For the nominal voltage conversion from 348 V on the PV side to 670 V on the DC-link side, the ideal boost-converter duty ratio is

$$D = 1 - \frac{V_{PV}}{V_{DC}}$$

which gives

$$D = 1 - \frac{348}{670} \approx 0.48$$

The allowable PV-side voltage ripple is selected as 1% of the nominal PV operating voltage. Therefore,

$$\Delta V_{PV} = 0.01 \times 348 = 3.48 \text{ V}$$

Using a boost-converter switching frequency of

$$f_s = 20 \text{ kHz},$$

the required input capacitance is calculated as

$$C_{in} = \frac{14.7 \times 0.48}{20 \times 10^3 \times 3.48}$$

$$C_{in} \approx 101.4\ \mu\text{F}$$

Accordingly, a nominal capacitance of

$$\boxed{C_{in} = 100\ \mu\text{F}}$$

is selected for the simulation model. This value is approximately equal to the calculated capacitance requirement and is used to limit the high-frequency voltage ripple at the PV terminals.

2.4 Boost Converter Design

The voltage available from the PV array is lower than the nominal DC-link voltage required by the motor-drive system. A boost converter is therefore employed to step up the PV array voltage and provide the required DC-link voltage while also enabling control of the PV operating point through variation of the converter duty ratio.

The boost converter is designed around the nominal maximum-power operating point of the PV array.

For an ideal boost converter operating in continuous conduction mode, the relationship between the input and output voltages is

$$V_{DC} = \frac{V_{PV}}{1 - D}$$

where D is the converter duty ratio. For the nominal voltage conversion from 348 V to 670 V, the required duty ratio is

$$D = 1 - \frac{348}{670} \approx 0.48$$

The boost-converter inductor is designed by limiting its peak-to-peak current ripple to 20% of the nominal input current. Since the average inductor current of an ideal boost converter is equal to the input current, the nominal inductor current is taken as the PV current at the maximum-power operating point. Therefore,

$$\Delta I_L = 0.2 I_{PV,mpp}$$

$$\Delta I_L = 0.2 \times 14.7 = 2.94 \text{ A}$$

The required inductance is determined from the inductor voltage during the switch ON interval:

$$L = \frac{V_{PV} D}{f_s \Delta I_L}$$

$$L \approx 2.84 \text{ mH}$$

A slightly higher practical value is therefore selected for the simulation model:

$$\boxed{L = 3 \text{ mH}}$$

The output of the boost converter is directly connected to the DC link, whose capacitance is designed separately based on the energy requirements of the motor-drive system.

2.5 DC-Link Capacitor Design

The DC-link capacitor is located between the boost converter and the three-phase voltage-source inverter and provides short-term energy buffering between the source and load sides of the system. During transient power imbalances, the capacitor can temporarily supply or absorb energy while limiting the resulting variation in the DC-link voltage.

The capacitor is sized using an energy-based approach. The energy stored in a capacitor at a voltage V is

$$E_C = \frac{1}{2} C_{DC} V^2$$

If the capacitor supplies a power P for a time interval Δt, while its voltage decreases from V_1 to V_2, the corresponding energy balance is

$$P\Delta t = \frac{1}{2} C_{DC}(V_1^2 - V_2^2)$$

Therefore, the required capacitance is

$$C_{DC} = \frac{2P\Delta t}{V_1^2 - V_2^2}.$$

The capacitor is designed for the high-speed operating region of the motor-pump system. For a centrifugal pump, the load torque varies approximately as the square of the mechanical speed:

$$T_L = K\omega_m^2$$

Consequently, the mechanical load power is

$$P_L = T_L\omega_m = K\omega_m^3$$

The power demand therefore increases rapidly with motor speed. At the same time, operation at higher motor speeds requires a higher stator voltage within the normal operating region of the drive, which in turn requires a sufficiently high DC-link voltage for the inverter. The high-speed operating condition is therefore associated with both a high power demand and a high DC-link voltage requirement. Accordingly, the nominal maximum design value of 670 V is considered for the capacitor-sizing calculation.

The capacitor is designed to provide temporary support at the rated power of

$$P = 4\ \text{kW}$$

for a selected interval of 50 boost-converter switching cycles. At a switching frequency of 20 kHz, the switching period is

$$T_s = \frac{1}{f_s} = \frac{1}{20 \times 10^3} = 50\ \mu\text{s}$$

Therefore, the selected energy-support interval is

$$\Delta t = 50T_s = 2.5\ \text{ms}$$

The allowable DC-link voltage reduction over this interval is selected as 2%. Taking the initial DC-link voltage as

$$V_1 = 670\ \text{V}$$

the minimum voltage after a 2% reduction is

$$V_2 = 0.98V_1 = 656.6\ \text{V}$$

The required DC-link capacitance is therefore

$$C_{DC} = \frac{2(4000)(2.5 \times 10^{-3})}{670^2 - 656.6^2}$$

$$C_{DC} \approx 1125\ \mu\text{F}$$

A capacitance value of

$$\boxed{C_{DC} = 1200\ \mu\text{F}}$$

is therefore selected for the simulation model. The selected capacitor provides short-term energy buffering between the PV-side converter and the motor drive while limiting the assumed DC-link voltage variation during transient power imbalance.

2.6 Three-Phase Voltage Source Inverter

The DC-link voltage is converted into a controlled three-phase AC supply for the induction motor using a two-level three-phase voltage-source inverter (VSI). The inverter is implemented using the Universal Bridge block in MATLAB/Simulink with six IGBT-diode switching devices.

The inverter is supplied directly from the DC link, while its six switching signals are generated using sinusoidal pulse-width modulation (SPWM). The three-phase reference voltages produced by the induction motor control system are compared with a high-frequency carrier signal to generate the required gate pulses.

The achievable fundamental output voltage of the inverter is limited by the available DC-link voltage and the modulation index. For linear SPWM operation, the maximum line-to-line RMS fundamental voltage is given by

$$V_{LL,\mathrm{rms}} = \frac{\sqrt{3}}{2\sqrt{2}} m_a V_{DC}$$

For the maximum linear modulation index $m_a = 1$, supplying the rated motor voltage of 400 V requires a theoretical minimum DC-link voltage of approximately

$$V_{DC,\min} = \frac{2\sqrt{2}}{\sqrt{3}}(400) \approx 653$$

The nominal DC-link design value of 670 V therefore provides sufficient voltage capability for generating the rated motor voltage while maintaining a small modulation margin.

2.7 Induction Motor Parameters

The motor used in the water-pumping system is a three-phase squirrel-cage induction motor rated at 4 kW, 400 V, and 50 Hz, with a rated speed of 1430 rpm. The machine is represented using the Asynchronous Machine SI Units block available in MATLAB/Simulink. The mechanical input of the machine is configured as load torque, allowing the centrifugal pump torque characteristic to be applied directly to the motor shaft.

The electrical parameters of the selected induction motor are a stator resistance of

$$R_s = 1.405\ \Omega$$

a rotor resistance referred to the stator of

$$R_r = 1.395\ \Omega$$

and equal stator and rotor leakage inductances of

$$L_{ls} = L_{lr} = 5.839\ \text{mH}$$

The mutual inductance of the machine is

$$L_m = 0.1722\ \text{H}$$

The mechanical parameters consist of a combined rotor inertia of

$$J = 0.0131\ \text{kg} \cdot \text{m}^2$$

and a viscous friction coefficient of

$$F = 0.002985\ \text{N} \cdot \text{m} \cdot \text{s}$$

The machine has two pole pairs, corresponding to a four-pole induction motor. These electrical and mechanical parameters are subsequently used in the development of the induction motor model and the IFOC control system.

2.8 Centrifugal Pump Load Modelling

The mechanical load connected to the induction motor is modelled as a centrifugal pump. Unlike a constant-torque load, the torque required by a centrifugal pump varies approximately with the square of its rotational speed. The load-torque characteristic is represented as

$$T_L = K\omega_m^2$$

where T_L is the pump load torque, ω_m is the mechanical rotor speed, and K is the pump load coefficient.

To ensure that the pump load is consistent with the power rating of the selected induction motor, the coefficient K is selected as

$$K = 0.0012$$

The resulting load-torque characteristic used in the simulation is therefore

$$\boxed{T_L = 0.0012\ \omega_m^2}$$

Since mechanical power is given by

$$P_L = T_L \omega_m$$

substitution of the centrifugal pump torque characteristic gives

$$P_L = K \omega_m^3$$

Therefore,

$$\boxed{P_L \propto \omega_m^3}$$

This cubic relationship indicates that the power required by the pump increases rapidly as the motor speed is increased. Consequently, relatively small increases in operating speed can result in significantly larger power demand from the PV-fed motor-drive system. This characteristic becomes particularly important when evaluating the maximum sustainable operating point of the standalone system.

In the Simulink implementation, the measured rotor speed is used to calculate the pump torque according to the quadratic torque-speed relationship. The absolute value of speed is included in the implementation so that the load torque retains the appropriate opposing characteristic with respect to the direction of rotation. The calculated load torque is then applied directly to the mechanical torque input of the induction machine.

3. Indirect Field-Oriented Control of the Induction Motor

The induction motor in the proposed water-pumping system is controlled using Indirect Field-Oriented Control (IFOC). The control system regulates the motor speed while independently controlling the flux-producing and torque-producing components of the stator current in a synchronously rotating reference frame.

The IFOC system consists of an speed control loop, generation of the d- and q-axis current references, calculation of the slip and synchronous electrical speeds, calculation of the corresponding d- and q-axis voltage references, and generation of the three-phase voltage references. The overall control sequence is developed in the following sections.

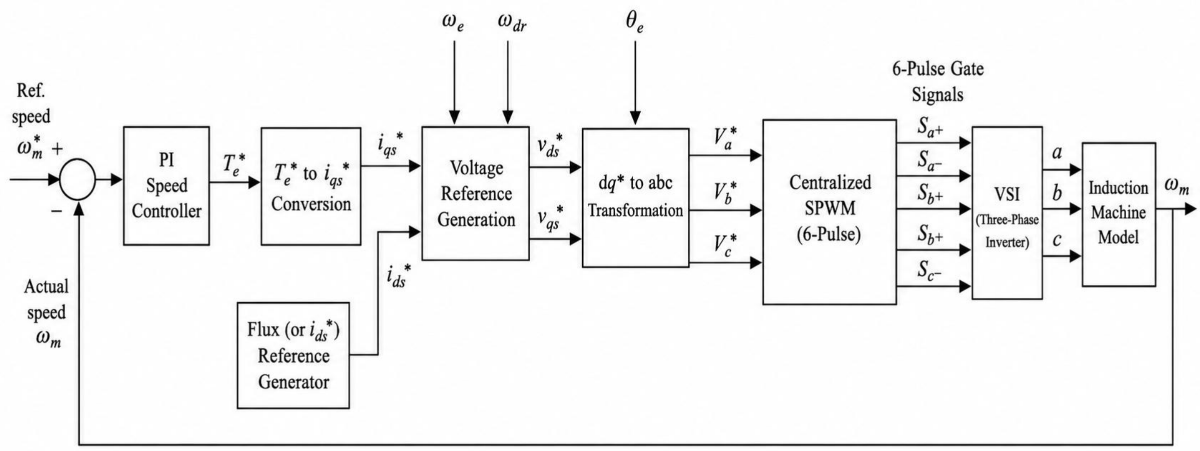

Figure 3.1: Overall Block Diagram of the Indirect Field-Oriented Control System

3.1 Principle of Indirect Field-Oriented Control

In a three-phase induction motor, the stator currents are inherently responsible for establishing both the magnetic flux and the electromagnetic torque. This coupling makes direct control of the motor more complex than that of a separately excited DC motor, where the field and armature currents can be independently controlled.

Field-oriented control addresses this problem by representing the three-phase motor quantities in a synchronously rotating d-q reference frame. By aligning the d-axis of this reference frame with the rotor-flux vector, the stator-current components can be separated according to their primary control functions. Under rotor-flux orientation, the d-axis current component, i_{ds} acts primarily as the flux-producing component, while the q-axis current component, i_{qs} acts primarily as the torque-producing component. The motor can therefore be controlled through two approximately decoupled current components.

The required torque-producing current is determined according to the demanded electromagnetic torque, while the flux-producing current is maintained at the value required to establish the desired rotor flux. The resulting control structure can be represented conceptually as

$$i_{ds}^{*} \rightarrow \text{Flux Control}$$

and

$$i_{qs}^{*} \rightarrow \text{Torque Control}$$

In indirect field-oriented control, the orientation of the rotating reference frame is obtained without directly measuring the rotor-flux position. Instead, the slip angular frequency is calculated from the motor parameters and the commanded d- and q-axis currents. The synchronous electrical angular speed is then obtained by combining the rotor electrical speed with the calculated slip speed:

$$\omega_e = \omega_r + \omega_{sl}$$

The corresponding synchronous electrical angle is obtained as

$$\theta_e = \int \omega_e \, dt$$

This angle is subsequently used in the reference-frame transformations required for the generation of the three-phase voltage references.

3.2 Current Control and Voltage Reference Generation

The implementation of IFOC requires the generation of the flux-producing d-axis current reference and the torque-producing q-axis current reference. In the developed control system, the d-axis current reference is determined from the rated rotor-flux requirement, while the q-axis current reference is calculated from the electromagnetic torque reference generated by the speed controller.

The rated rotor flux is first determined from the rated induced voltage of the machine using the RMS voltage relation

$$E_{\text{rms}} = \sqrt{2}\pi K_w f_{me} \lambda_{\text{r,rated}}$$

where E_{rms} is the rated induced phase voltage, K_w is the winding factor, f_{me} is the rated electrical frequency, and λ_{rated} is the required rated rotor flux. The corresponding d-axis current reference is then obtained from the rotor-flux relationship

$$\lambda_{\text{r,rated}} = \lambda_{dr} = L_m i_{ds}^*$$

Therefore

$$i_{ds}^* = \frac{\lambda_r}{L_m}$$

For the selected induction motor parameters, the resulting constant flux-producing current reference is

$$i_{ds}^* = 6.037 \text{ A}$$

This value corresponds to the rated rotor-flux requirement and is maintained as the d-axis current reference throughout the considered operating range. Although the rated voltage and frequency are used to determine this reference, the motor is not continuously operated at rated voltage. Maintaining i_{ds}^* approximately constant maintains the desired rotor flux, while the voltage generated by the current-control system varies according to the motor operating speed and load.

The torque-producing current reference is obtained from the output of the speed-control loop. The reference speed is compared with the measured mechanical rotor speed, and the resulting speed error is supplied to a PI controller

$$e_\omega = \omega_m^* - \omega_m$$

The output of the speed PI controller provides the required electromagnetic torque reference T_e^*

Under rotor-flux-oriented operation, the electromagnetic torque is expressed in terms of the d- and q-axis current components as

$$T_e = \left(\frac{3}{2}\frac{P}{2}\frac{L_m^2}{L_r'}\right) i_{ds} i_{qs}$$

where P is the number of machine poles and L_r' is the total rotor inductance referred to the stator. The total rotor inductance is obtained as

$$L_r' = L_m + L_{lr}'$$

Since i_{ds}^* is maintained constant, the required torque-producing current reference can be calculated directly from the torque reference as

$$i_{qs}^{*} = \frac{T_e^{*}}{\left(\frac{3}{2}\frac{P}{2}\frac{L_m^2}{L_r'}\right) i_{ds}^{*}}$$

Thus, no additional controller is required for generating the q-axis current reference. The single speed PI controller generates T_e^{*}, which is then converted algebraically into i_{qs}^{*}.

The synchronous reference-frame angle is generated indirectly using the rotor electrical speed and the calculated slip angular frequency. The measured mechanical rotor speed is converted to electrical rotor speed as

$$\omega_r = \frac{P}{2}\omega_m$$

The slip angular frequency is calculated using the current references and rotor parameters

$$\omega_{sl} = \frac{R_r'}{L_r'}\frac{i_{qs}^{*}}{i_{ds}^{*}}$$

The synchronous electrical angular speed is then obtained as

$$\omega_e = \omega_r + \omega_{sl}$$

Finally, the synchronous electrical angle used for the reference-frame transformations is obtained by integration

$$\theta_e = \int \omega_e \, dt$$

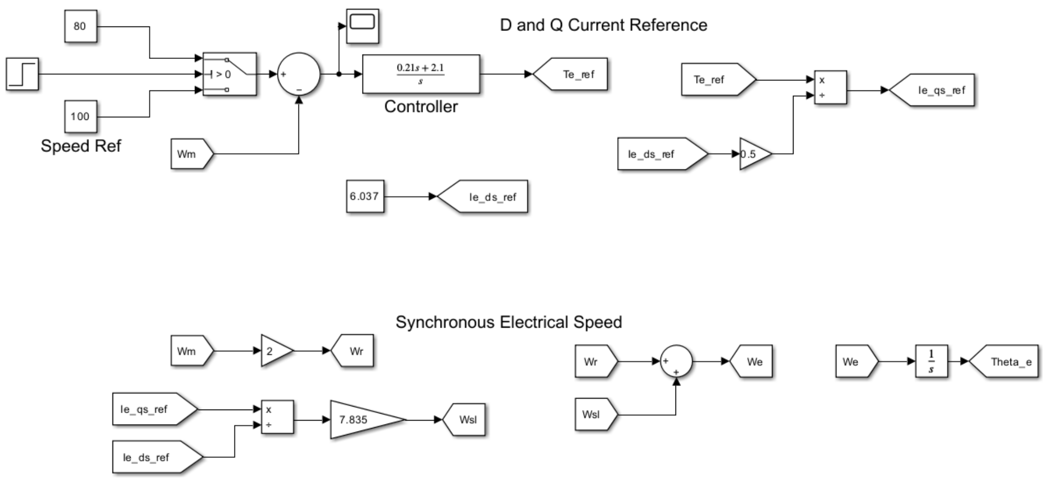

Figure 3.2: Simulink Implementation of Current Reference and Synchronous Angle Generation

The generated i_{ds}^* and i_{qs}^*, together with the synchronous electrical angular speed ω_e and the rotor-flux linkage ψ_{dr}, are used to calculate the required d- and q-axis voltage references v_{ds}^* and v_{qs}^*. These voltage references are subsequently transformed into the three-phase voltage references v_a^*, v_b^*, and v_c^* using the synchronous electrical angle θ_e. The resulting three-phase voltage references are then supplied to the inverter modulation stage for generation of the required switching signals.

3.3 Voltage Reference Generation and Reference-Frame Transformation

Once the d- and q-axis current references have been generated, the corresponding voltage references required for motor operation are calculated using the induction motor voltage equations in the synchronously rotating reference frame. The calculation uses the reference current components, synchronous electrical angular speed, machine parameters, and rotor-flux linkage.

Under rotor-flux-oriented operation, the rotor flux is aligned with the d-axis. The implemented voltage-reference equations are

$$v_{ds}^* = R_s i_{ds}^* + \sigma L_s \frac{di_{ds}^*}{dt} - \omega_e \sigma L_s i_{qs}^*$$

$$v_{qs}^* = R_s i_{qs}^* + \sigma L_s \frac{di_{qs}^*}{dt} + \omega_e \sigma L_s i_{ds}^* + \omega_e \frac{L_m}{L_r'} \psi_{dr}$$

where R_s is the stator resistance, σL_s represents the stator transient inductance, ω_e is the synchronous electrical angular speed, and ψ_{dr} is the d-axis rotor-flux linkage.

Since the d-axis current reference is maintained constant for constant-flux operation

$$\frac{di_{ds}^*}{dt} = 0$$

the d-axis voltage-reference equation reduces to

$$v_{ds}^* = R_s i_{ds}^* - \omega_e \sigma L_s i_{qs}^*$$

The calculated d- and q-axis voltage references are expressed in the synchronously rotating reference frame and must therefore be transformed into three-phase voltage references for the inverter. In the developed model, this conversion is performed directly using the inverse dq-to-abc transformation

$$
\begin{bmatrix} v_a^* \\ v_b^* \\ v_c^* \end{bmatrix} = \sqrt{\frac{2}{3}} \begin{bmatrix} \cos\theta_e & -\sin\theta_e \\ \cos\left(\theta_e - \frac{2\pi}{3}\right) & -\sin\left(\theta_e - \frac{2\pi}{3}\right) \\ \cos\left(\theta_e + \frac{2\pi}{3}\right) & -\sin\left(\theta_e + \frac{2\pi}{3}\right) \end{bmatrix} \begin{bmatrix} v_{ds}^* \\ v_{qs}^* \end{bmatrix}
$$

where θ_e is the synchronous electrical angle calculated in the preceding section.

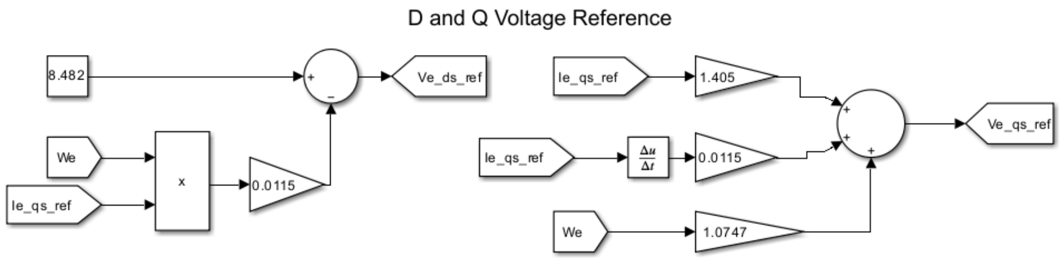

Figure 3.3: Simulink Implementation of d-q Voltage Reference Generation

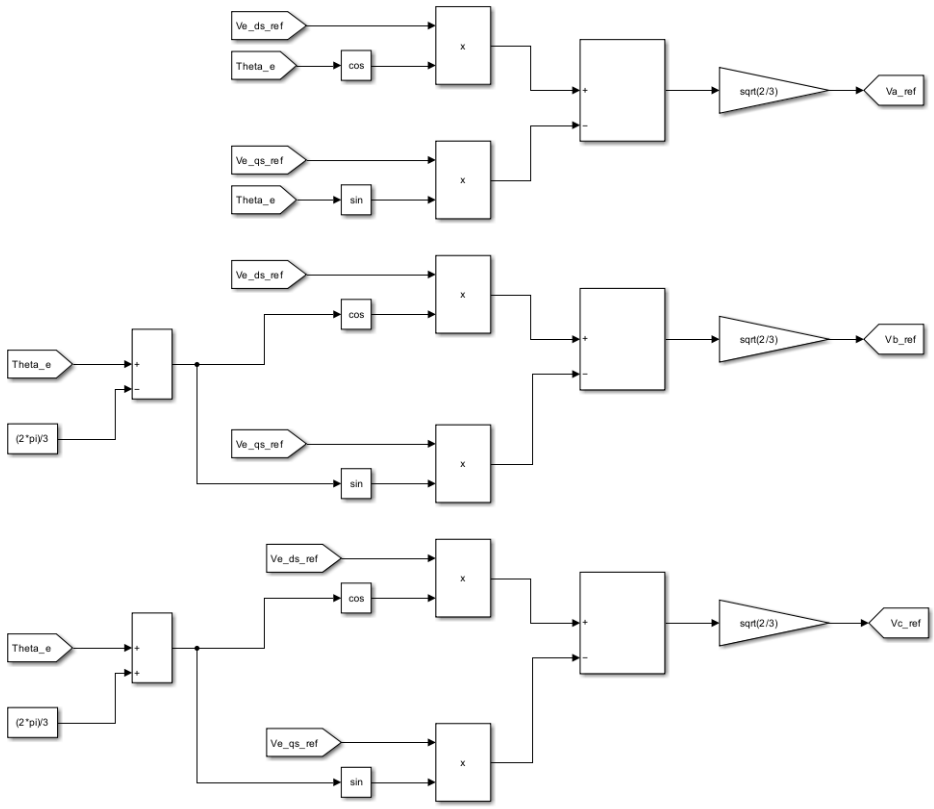

Figure 3.4: Simulink Implementation of Direct dq-to-abc Voltage Transformation

The resulting three-phase voltage references v_a^*, v_b^*, and v_c^* are subsequently supplied to the inverter modulation stage for generation of the required switching signals.

3.4 Speed Controller Design

The speed controller forms the outer control loop of the implemented IFOC system and generates the electromagnetic torque reference T_e^* from the difference between the reference and measured motor speeds. The controller was designed using the Bode plot loop-shaping approach based on the mechanical dynamics of the induction motor.

The mechanical dynamics of the motor are described by

$$J\frac{d\omega_m}{dt} + B\omega_m = T_e - T_L$$

For the purpose of controller design, the load torque is treated as an external disturbance. The transfer function between electromagnetic torque and mechanical rotor speed is therefore

$$G_m(s) = \frac{\omega_m(s)}{T_e(s)} = \frac{1}{Js + B}$$

Using the motor parameters

$$J = 0.0131\ \mathrm{kg \cdot m^2}$$
$$B = 0.002985\ \mathrm{N \cdot m \cdot s}$$

the mechanical plant can be expressed in normalized first-order form as

$$G_m(s) = \frac{1/B}{1 + (J/B)s}$$

which gives

$$G_m(s) = \frac{335}{1 + s/0.228}$$

A PI controller was selected to provide zero steady-state error for a constant speed reference while allowing the transient response and stability margins to be shaped in the frequency domain. The controller has the form

$$G_c(s) = K_p + \frac{K_i}{s}$$

where the controller zero is given by

$$\omega_z = \frac{K_i}{K_p}$$

The controller was designed through Bode plot loop shaping, with a target settling time of approximately 0.25s considered while selecting the desired loop bandwidth. The PI-controller zero was placed at

$$\omega_z = 10 \text{ rad/s}$$

The proportional gain was then adjusted through the open-loop Bode plot to obtain the desired crossover frequency and an adequate phase margin. The resulting controller gains were

$$K_p = 0.21$$
$$K_i = 2.1$$

The final PI speed controller is therefore

$$G_c(s) = \frac{0.21s + 2.1}{s}$$

The compensated open-loop system obtained from the loop-shaping design provided an infinite gain margin and a phase margin of approximately

$$PM = 62.04°$$

The corresponding gain crossover frequency was

$$\omega_{gc} = 18.28 \text{ rad/s}$$

The closed-loop response was subsequently evaluated using a unit-step input. The obtained response characteristics were a rise time of approximately 0.070s, a settling time of 0.379s, and a maximum overshoot of 22.61%. The peak response occurred at approximately 0.181s.

Following the controller design, the complete IFOC motor-drive system was initially tested using an ideal 700 V DC source feeding the three-phase VSI. The drive successfully tracked the commanded motor speeds, including operation up to 130 rad/s, confirming the satisfactory operation of the developed motor-control system before integration with the PV source and boost converter. This standalone test also provides a useful reference for the subsequent analysis of the operating limitations observed in the complete PV-fed system.

4. PV-Side Converter Control and Supervisory Strategy

The boost converter forms the interface between the PV array and the DC link of the motor-drive system. Unlike an ideal DC source, the PV array has a nonlinear voltage-current characteristic and a finite available power that varies with environmental conditions. The operating point imposed by the boost converter therefore directly influences both the power extracted from the PV array and the voltage available at the DC link.

In the developed system, the boost converter is required to satisfy two different operating objectives. Under normal conditions, the PV array is operated near its maximum power point to maximize the utilization of the available solar energy. However, the DC-link voltage must also remain sufficiently high to generate the voltage references demanded by the IFOC-controlled motor drive.

A supervisory control strategy is therefore implemented to coordinate maximum-power operation with the instantaneous DC-link voltage requirement of the motor drive. The following sections describe the converter control model, individual control objectives, minimum DC-link voltage calculation, and the mode-selection mechanism used to determine the final boost-converter duty ratio.

4.1 Boost Converter Control Model

The boost converter is controlled according to two different operating objectives in the developed system. During maximum-power operation, the converter regulates the PV-array voltage, whereas during DC-link-support operation, it regulates the output DC-link voltage. Since the controlled variable differs between these two operating modes, separate small-signal converter models were used for the design of the corresponding controllers.

The nominal converter operating point was selected as

$$V_{PV} = 348 \text{ V}$$

$$V_{DC} = 670 \text{ V}$$

The corresponding nominal duty ratio is

$$D = 1 - \frac{V_{PV}}{V_{DC}} \approx 0.48$$

For the PV-side voltage-control loop, the PV array is represented locally by its incremental resistance r_{pv} around the selected operating point. Using the small-signal converter equations and eliminating the inductor-current perturbation gives the duty-to-PV-voltage transfer function

$$\frac{\hat{V}_{pv}(s)}{\hat{d}(s)} = -\frac{V_{dc}}{LC_{in}s^2 + \frac{L}{r_{pv}}s + 1}$$

The parameters used for the input-side model are

$$L = 3\ \text{mH}$$
$$C_{in} = 100\ \mu\text{F}$$
$$V_{dc} = 670\ \text{V}$$
$$r_{pv} = 25\ \Omega$$

The negative sign in the transfer function represents the inverse relationship between the boost-converter duty ratio and the PV operating voltage. An increase in duty ratio causes the converter to draw the PV source more strongly, resulting in a reduction in the PV-array voltage. This transfer function is therefore used as the plant model for the PV-voltage controller associated with maximum-power-point operation.

For the DC-link voltage-control loop, the classical continuous-conduction-mode boost-converter control-to-output transfer function was used. The model is represented in canonical form as

$$G_{vd}(s) = \frac{\hat{V}_{dc}(s)}{\hat{d}(s)} = G_{d0}\frac{1 - \frac{s}{\omega_{\text{RHP}}}}{1 + \frac{s}{Q\omega_0} + \left(\frac{s}{\omega_0}\right)^2}$$

The output-side converter model contains a right-half-plane zero and therefore exhibits non-minimum-phase behaviour. An increase in duty ratio initially increases the time for which the inductor is disconnected from the output, causing the DC-link response to move temporarily in the opposite direction before the additional stored inductor energy is transferred to the output. This characteristic places an upper limit on the practical bandwidth of the DC-link voltage-control loop.

For the designed converter, with the boost inductance and DC-link capacitance given by

$$L = 3\ \text{mH}$$
$$C_{DC} = 1200\ \mu\text{F}$$

the parameters of the output-side small-signal model were obtained as

$$G_{d0} = 1291$$
$$Q = 36.8$$
$$\omega_0 = 2898\ \text{rad/s}$$
$$\omega_{\text{RHP}} = 10000\ \text{rad/s}$$

The resulting duty-to-DC-link-voltage transfer function is therefore

$$G_{vd}(s) = 1291 \frac{1 - \frac{s}{10000}}{1 + \frac{s}{(36.8)(2898)} + \left(\frac{s}{2898}\right)^2}$$

The two developed small-signal models represent the different dynamics encountered by the boost converter under the two control objectives. The input-side duty-to-PV-voltage model is used for the design of the maximum-power operating controller, while the output-side duty-to-DC-link-voltage model is used for the design of the DC-link-support controller.

4.2 Maximum Power Point Voltage Control

The first operating objective of the boost-converter control system is to maintain the PV array near its maximum power point. For the selected PV-array configuration, the reference maximum-power-point voltage is approximately

$$V_{MPP}^{*} = 348 \text{ V}$$

The measured PV voltage is compared with this reference, and the resulting voltage error is processed by the PV-voltage controller. The controller output determines the duty-ratio command associated with maximum-power operation

$$e_{PV} = V_{PV} - V_{MPP}^{*}$$
$$e_{PV} \rightarrow C_{PV}(s) \rightarrow D_{MPP}$$

The error polarity is selected according to the negative duty-to-PV-voltage relationship of the boost converter established in Section 4.1. An increase in duty ratio causes the PV voltage to decrease, and therefore the controller sign must account for this inverse plant behaviour.

The PV-voltage controller was initially designed using Bode-plot loop shaping based on the small-signal duty-to-PV-voltage transfer function. The initial loop-shaping process indicated that integral action was sufficient to provide the required low-frequency gain and eliminate steady-state voltage error. An integral controller was therefore initially considered in the form

$$C_{PV}(s) = \frac{K_i}{s}$$

The controller parameters were subsequently refined through tuning of the closed-loop response. A small proportional component was introduced to improve the practical transient response and robustness of the voltage-control loop while retaining the dominant integral action identified through loop shaping.

The final controller gains were selected as

$$K_p = 1 \times 10^{-4}$$
$$K_i = 0.3$$

The resulting PV-voltage PI controller is therefore

$$C_{PV}(s) = 10^{-4} + \frac{0.3}{s}$$

The compensated open-loop system provided a gain margin of approximately 7.22dB and a phase margin of approximately 92.47°. The gain crossover frequency was approximately 203.95 rad/s.

The corresponding closed-loop response exhibited a rise time of approximately 10.72ms, a settling time of 21.34ms, and an overshoot of approximately 0.193%. These results indicate a fast and well-damped PV-voltage response with adequate stability margins.

The resulting duty-ratio command D_{MPP} is used when the supervisory controller selects maximum-power operation. Under this operating mode, regulation of the PV-array voltage around V_{MPP}^* maintains the array near its maximum-power operating region. The interaction between this operating objective and the motor-drive DC-link voltage requirement is addressed in the subsequent sections.

4.3 DC-Link Voltage Control

The second operating objective of the boost-converter control system is to maintain the DC-link voltage at the level required by the motor drive. A separate DC-link voltage controller was therefore designed using the duty-to-output-voltage small-signal model developed in Section 4.1.

The controller design was performed around the nominal converter operating point corresponding to

$$V_{DC} = 670 \text{ V}$$
$$V_{PV} = 348 \text{ V}$$
$$P = 4 \text{ kW}$$

The nominal duty ratio is

$$D = 1 - \frac{348}{670} = 0.4806$$

For the small-signal converter model, the inverter and motor-drive system were represented by an equivalent resistive load at the nominal operating point. The equivalent resistance is obtained as

$$R = \frac{V_{DC}^2}{P}$$

which gives

$$R = \frac{670^2}{4000} = 112.225\,\Omega$$

Using the converter parameters

$$L = 3\text{ mH}$$

$$C_{DC} = 1200\,\mu\text{F}$$

the DC gain of the control-to-output transfer function is obtained as

$$G_{d0} = \frac{V_{DC}}{1-D}$$

$$G_{d0} \approx 1290$$

The natural frequency is calculated as

$$\omega_0 = (1-D)\sqrt{\frac{R}{LC_{DC}}}$$

resulting in

$$\omega_0 \approx 2898\text{ rad/s}$$

The quality factor is given by

$$Q = R(1-D)\sqrt{\frac{C_{DC}}{L}}$$

which gives

$$Q \approx 36.8$$

The right-half-plane zero is located at approximately

$$\omega_{\text{RHP}} = \frac{R(1-D)^2}{L}$$

$$\omega_{\text{RHP}} \approx 10090\text{ rad/s}$$

which was approximated as 10000rad/s in the converter model.

The resulting control-to-output transfer function used for controller design was therefore

$$G_{vd}(s) = 1291\frac{1-\frac{s}{10000}}{1+\frac{s}{(36.8)(2898)}+\left(\frac{s}{2898}\right)^2}$$

The DC-link voltage controller was initially designed using Bode-plot loop shaping. The design indicated that integral action was sufficient to obtain the required low-frequency gain and eliminate steady-state voltage error while maintaining the crossover frequency well below the converter resonance and the right-half-plane zero. An integral controller with

$$K_i = 0.058$$

was therefore obtained as the initial design.

The controller was subsequently tuned for implementation in the complete switching model. Increasing the proportional gain was found to reduce the robustness of the control loop by shifting its dynamics toward the highly resonant region of the boost-converter plant. Consequently, only a very small proportional component was retained in the final controller.

The final implemented gains were

$$K_p = 1 \times 10^{-6}$$

$$K_i = 0.04$$

giving the DC-link voltage controller

$$C_{DC}(s) = 10^{-6} + \frac{0.04}{s}$$

The controller therefore behaves predominantly as an integral controller over the relevant control bandwidth, while retaining a small proportional component in the implemented PI structure. The resulting controller output forms the duty-ratio command associated with DC-link voltage support.

4.4 Minimum DC-Link Voltage Requirement

The voltage required at the DC link varies with the operating condition of the induction motor. At lower motor speeds, the voltage references generated by the IFOC system are relatively small, whereas an increase in motor speed generally requires larger stator voltage references. Maintaining a fixed high DC-link voltage under all operating conditions is therefore not always necessary.

The IFOC system developed in Chapter 3 generates the instantaneous three-phase voltage references

$$v_a^* \ v_b^* \ v_c^*$$

For a two-level three-phase VSI using sinusoidal PWM, the maximum phase-voltage reference that can be synthesized is related to the available DC-link voltage and modulation index by

$$V_{\text{phase,peak}} = \frac{m_a V_{DC}}{2}$$

Therefore, the DC-link voltage required to generate a particular instantaneous phase-voltage command can be obtained as

$$V_{DC} = \frac{2 \mid v_{\text{phase}}^* \mid}{m_a}$$

Since all three phase-voltage references must remain within the available modulation range, the largest instantaneous magnitude among the three references determines the minimum required DC-link voltage. The required voltage is therefore calculated as

$$V_{DC,\min} = \frac{2\max(|\ v_a^*\ |,|\ v_b^*\ |,|\ v_c^*\ |)}{m_a}$$

where m_a represents the maximum allowable modulation index used in the inverter modulation stage.

This approach allows the minimum DC-link voltage requirement to be determined directly from the instantaneous voltage demand of the motor-control system rather than maintaining a fixed DC-link reference irrespective of the operating condition.

The calculated $V_{DC,\min}$ is subsequently used by the supervisory control system to evaluate whether the available DC-link voltage is sufficient for the motor drive. When sufficient voltage margin is available, the boost converter can prioritize operation of the PV array near its maximum-power point. When the DC-link voltage approaches the minimum required level, the control objective can instead be shifted toward DC-link voltage support.

4.5 Supervisory Mode Selection and Hysteresis

The developed boost-converter control system contains two operating modes. The first mode prioritizes the minimum DC-link voltage required by the motor drive, while the second mode operates the PV array at its maximum power point. A supervisory controller determines the active operating mode according to the relationship between the measured DC-link voltage and the dynamically calculated minimum DC-link voltage.

The two operating modes are defined as

$$\text{Mode} = 0 \ \text{DC-link voltage regulation}$$

$$\text{Mode} = 1 \ \text{Maximum-power-point operation}$$

The system is initialized in DC-link regulation mode to ensure that the voltage requirement of the motor drive is given priority during startup.

A direct comparison between V_{DC} and $V_{DC,\min}$ was initially found to cause repeated switching when the DC-link voltage remained close to the mode-transition boundary. To prevent this behaviour, a persistent-state hysteresis mechanism was introduced. The hysteresis factor used in the final implementation is

$$H = 0.02$$

When the system is operating in DC-link regulation mode, it remains in this mode until the measured DC-link voltage exceeds the minimum requirement by the specified hysteresis margin

$$V_{DC} > (1 + H)V_{DC,\min}$$

Once this condition is satisfied, the supervisory controller switches to maximum-power-point operation.

When operating in maximum-power-point mode, the system remains in this mode as long as the DC-link voltage remains above the minimum required value. If the voltage falls below the required level

$$V_{DC} < V_{DC,\min}$$

the supervisory controller immediately returns to DC-link regulation mode.

The resulting mode-selection logic can therefore be summarized as

$$\text{Mode} = 0 \overset{V_{DC}>(1+H)V_{DC,\min}}{\rightarrow} \text{Mode} = 1$$

$$\text{Mode} = 1 \overset{V_{DC}<V_{DC,\min}}{\rightarrow} \text{Mode} = 0$$

The use of separate transition thresholds prevents rapid switching between the two operating modes near the boundary while ensuring that DC-link voltage support is restored whenever the available voltage becomes insufficient for the motor drive.

Once the operating mode has been selected, a second supervisory block generates the corresponding voltage reference for the active control objective.

In Mode 0, the minimum DC-link voltage calculated from the motor-side voltage references is directly assigned as the DC-link voltage reference

$$V_{DC}^{*} = V_{DC,\min}$$

The DC-link voltage controller then regulates the boost converter according to this dynamically varying reference.

In Mode 1, the PV-voltage reference is generated using the Incremental Conductance maximum-power-point tracking algorithm. The algorithm uses the measured PV voltage and current and evaluates the incremental and instantaneous conductances.

At the maximum power point

$$\frac{dP}{dV} = 0$$

Since

$$P = VI$$

the maximum-power condition becomes

$$\frac{dI}{dV} = -\frac{I}{V}$$

The implemented algorithm compares the incremental conductance

$$\frac{\Delta I}{\Delta V}$$

with the negative instantaneous conductance

$$-\frac{I_{PV}}{V_{PV}}$$

and adjusts the PV-voltage reference accordingly. The reference is increased or decreased in discrete steps of

$$\Delta V_{ref} = 1\ \text{V}$$

depending on the location of the present operating point relative to the maximum power point. The reference is initialized at

$$V_{PV}^{*} = 348\ \text{V}$$

and constrained within the selected operating limits

$$200\ \text{V} \leq V_{PV}^{*} \leq 430\ \text{V}$$

The generated voltage reference is then supplied to the corresponding voltage controller according to the selected operating mode.

Thus, the supervisory strategy coordinates the motor-drive voltage requirement with maximum solar-power extraction. DC-link regulation is given priority whenever the available DC-link voltage approaches the minimum required level, while Incremental Conductance MPPT is activated when sufficient DC-link voltage margin is available.

4.6 Controller Switching and Duty-Ratio Generation

The PV-voltage and DC-link voltage controllers generate separate duty-ratio commands corresponding to their respective control objectives. Since only one control objective is active at a given time, the supervisory mode signal is used to enable the appropriate controller and select the corresponding duty-ratio command.

The control selection is defined as

$$\text{Mode} = 0 \rightarrow D^{*} = D_{DC}$$
$$\text{Mode} = 1 \rightarrow D^{*} = D_{PV}$$

During the initial implementation, the inactive controller continued to accumulate its control error even though its output was not being applied to the boost converter. Consequently, the internal integrator state could become excessively large while the controller remained inactive. When the operating mode subsequently changed, the stored integrator state produced an abrupt duty-ratio command and degraded the transition between the two control modes.

To avoid this behaviour, the two controllers were provided with mode-dependent enable and reset logic. The DC-link controller is activated during Mode 0, while the PV-voltage controller is activated during Mode 1. The inactive controller is prevented from accumulating uncontrolled integral action, ensuring that its internal state does not produce a large transient when the supervisory system changes operating modes.

The corresponding active controller output is selected using the supervisory mode signal

$$D^* = \begin{cases} D_{DC}, & \text{Mode} = 0 \\ D_{PV}, & \text{Mode} = 1 \end{cases}$$

The selected duty-ratio command is subsequently passed through a saturation block. The maximum allowable duty ratio was limited to

$$D_{\max} = 0.95$$

This limit prevents the boost converter from being commanded toward an unrealizable duty ratio approaching unity and provides a practical upper bound for converter operation.

The final duty-ratio command is therefore constrained as

$$0 \leq D \leq 0.95$$

The resulting command is then used for PWM generation of the boost-converter switching signal.

This completes the PV-side control architecture. The supervisory controller first determines whether maximum-power operation or DC-link voltage support should be prioritized, activates the corresponding voltage-control path, and selects its duty-ratio command for operation of the boost converter.

5. Simulation Results and System Performance

5.1 Steady-State System Performance

The complete PV-fed induction motor drive was first evaluated under constant environmental and operating conditions to verify the steady-state performance of the integrated system. The PV array was operated at an irradiance of $1000\ \mathrm{W/m^2}$and a temperature of $25°\mathrm{C}$, while the motor speed reference was maintained at

$$\omega_m^* = 100\ \mathrm{rad/s}$$

Under these conditions, the supervisory control system coordinated the PV-side boost converter with the voltage requirement of the motor drive. The motor accelerated from standstill and reached the commanded operating speed, demonstrating successful integration of the PV source, boost converter, DC link, VSI, IFOC system, induction motor, and centrifugal pump load.

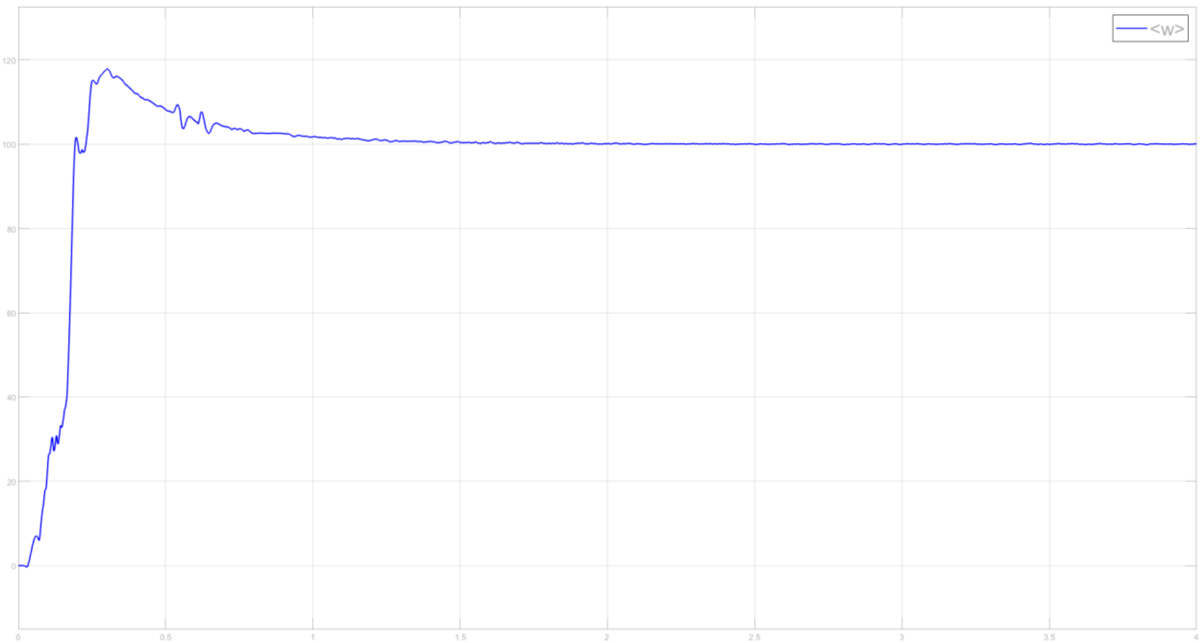

Figure 5.1: Motor Speed Under Steady Operating Conditions

The actual motor speed converged to the commanded reference and remained stable after the initial transient. This confirms that the developed IFOC system was able to provide the electromagnetic torque required for acceleration and subsequently maintain the desired operating speed against the centrifugal pump load.

The PV-side and DC-link quantities were monitored to evaluate the operation of the power-conversion stage. The boost converter maintained the required energy transfer from the PV array while the DC-link voltage remained sufficient to support the voltage demand of the motor drive.

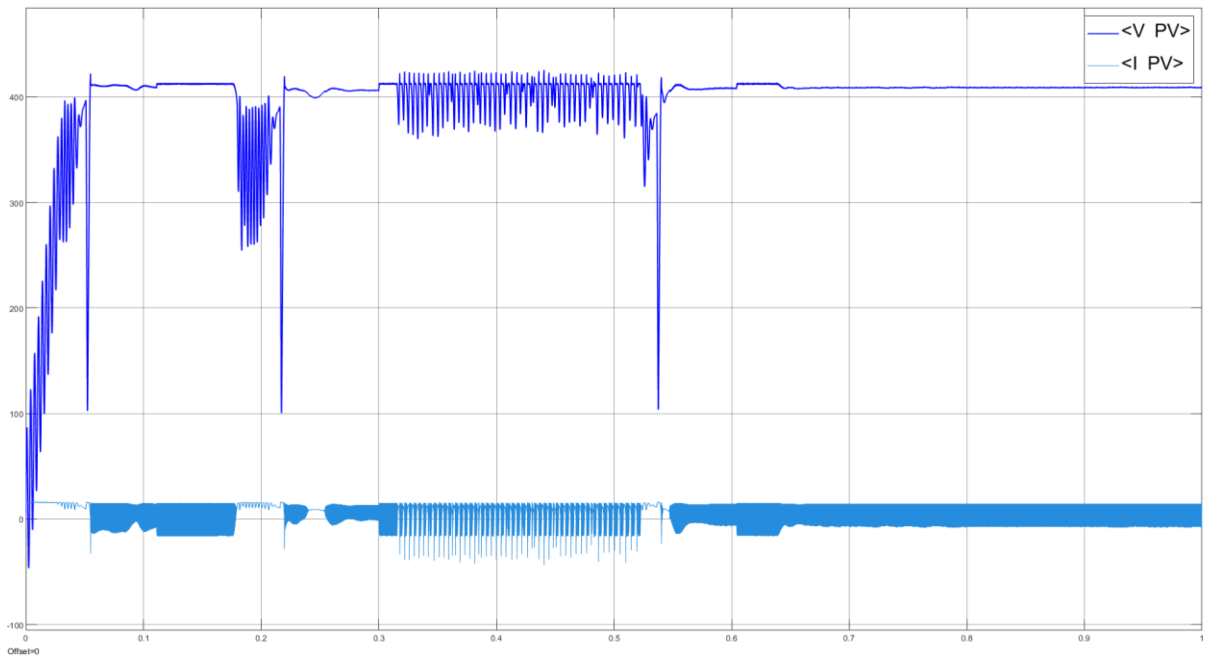

Figure 5.2: PV Voltage and PV Current During Steady-State Operation

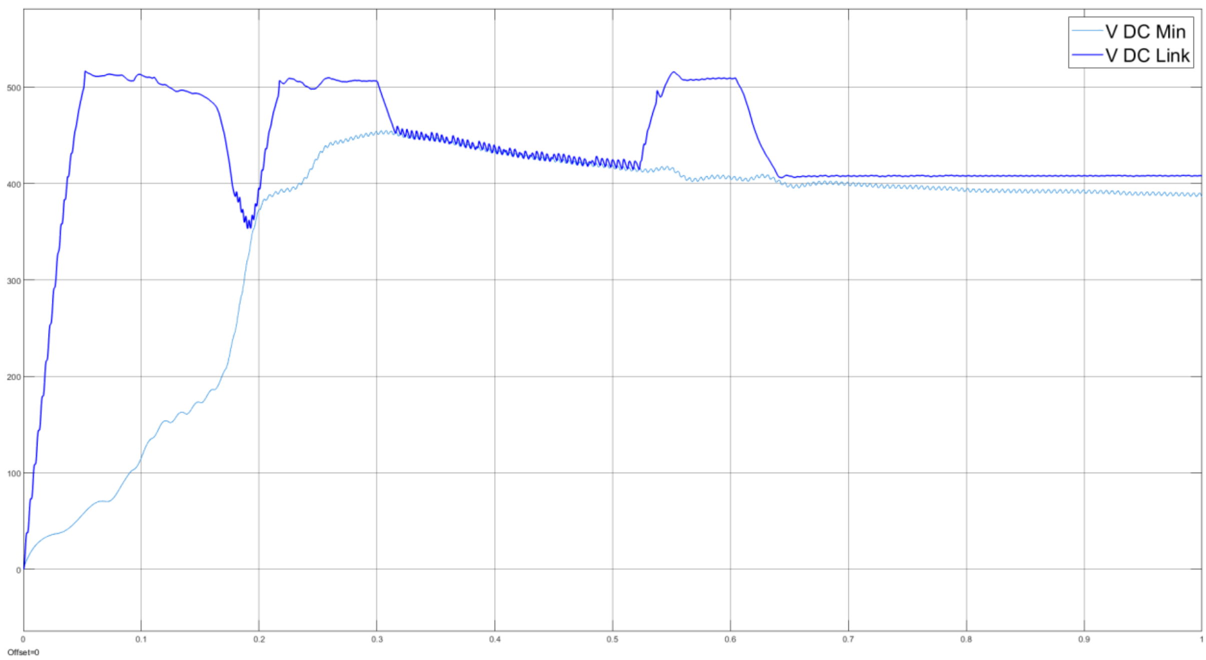

Figure 5.3: DC-Link Voltage and Minimum Required DC-Link Voltage

Overall, the steady-state simulation demonstrates that the complete system can operate successfully at the selected feasible operating point while supplying the centrifugal pump load and maintaining the motor at the commanded speed. The response of the integrated system to changes in environmental conditions and speed demand is examined in the following section.

5.2 Performance Under Environmental and Speed-Reference Variations

The dynamic performance of the complete system was evaluated by subjecting it to variations in solar irradiance, operating temperature, and motor speed reference within a single simulation. This test was performed to examine the ability of the integrated control system to maintain stable motor operation while both the available PV power and the required motor operating point were varied.

The system was initially operated at an irradiance of $1000\ \mathrm{W/m^2}$, a temperature of $25°\mathrm{C}$, and a motor speed reference of 100 rad/s.

At $t = 3$ s, the irradiance was reduced from

$$1000\ \mathrm{W/m^2} \rightarrow 700\ \mathrm{W/m^2}$$

At $t = 5$ s, the PV-cell temperature was increased from

$$25°\mathrm{C} \rightarrow 30°\mathrm{C}$$

Finally, at $t = 7$ s, the motor speed reference was reduced from

$$100\ \mathrm{rad/s} \rightarrow 80\ \mathrm{rad/s}$$

This sequence allows the response of the integrated system to source-side environmental variations and a subsequent change in motor operating demand to be evaluated within a single simulation.

These variations represent simultaneous changes in both the available source power and the power demanded by the motor-pump system.

The motor maintained operation close to the commanded speed during the environmental variations, demonstrating that the system remained stable despite changes in the PV-array operating characteristics and available solar power. Following the reduction in speed reference, the motor successfully transitioned to the new operating point and settled near the commanded speed.

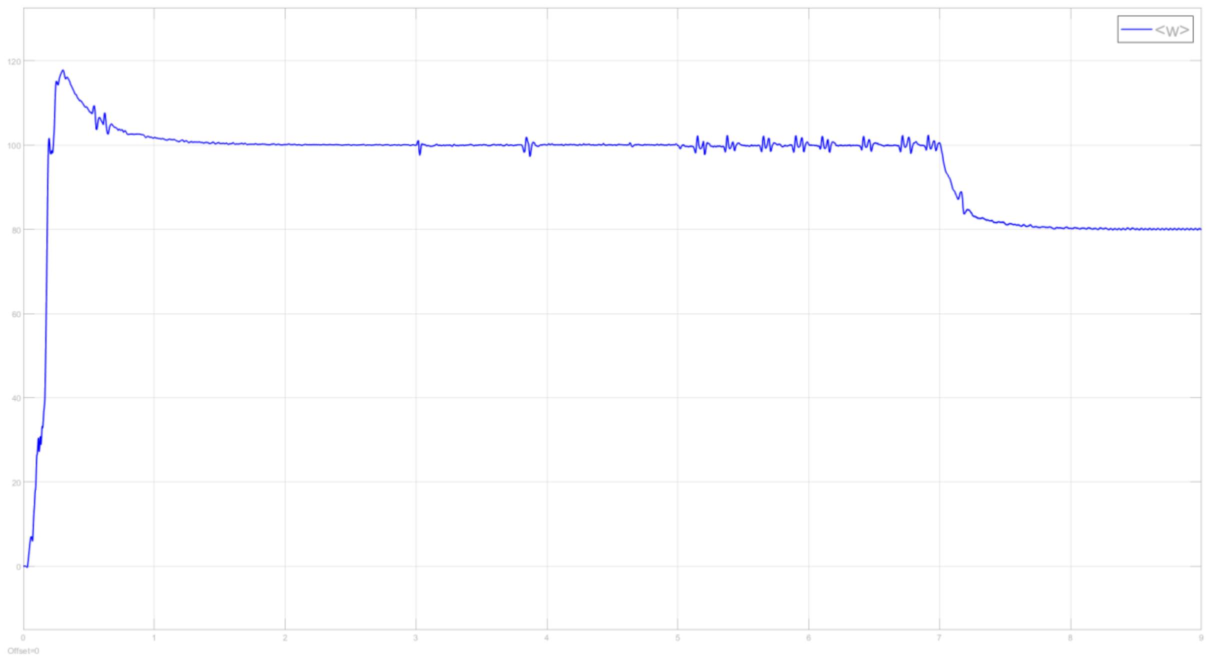

Figure 5.4: Reference and Actual Motor Speed Under Environmental and Speed-Reference Variations

The DC-link response was monitored together with the dynamically calculated minimum voltage requirement. Throughout the feasible operating conditions considered in this test, the available DC-link voltage remained sufficient to support the motor-drive voltage demand.

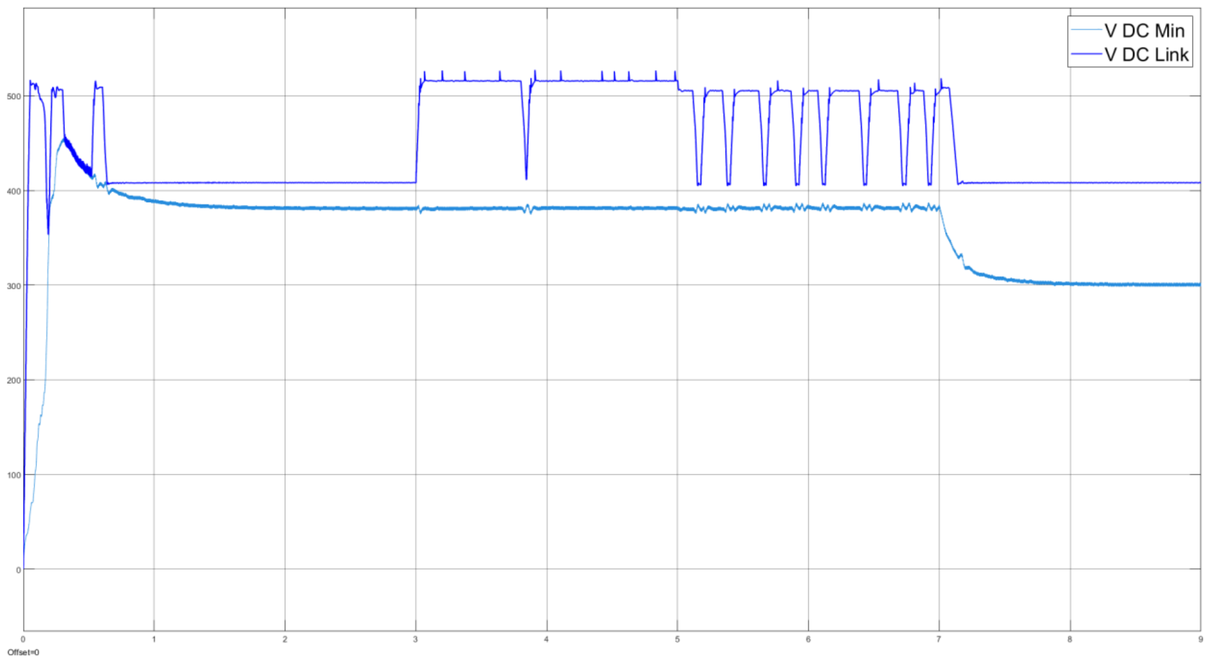

Figure 5.5: DC-Link Voltage and Minimum Required DC-Link Voltage During the Combined Dynamic Test

The combined disturbance test demonstrates that the developed system can maintain stable operation under variations in environmental conditions and motor speed demand. However, this successful operation is dependent on the required load power remaining within the sustainable power capability of the standalone PV source. The operating boundary associated with this limitation is investigated separately in the following chapter.

6. Operating Limit and Source-Side Collapse Analysis

During the simulation of the complete standalone PV-fed induction motor pumping system, it was observed that stable operation could be maintained only up to a certain motor speed. Beyond this operating point, the PV-side voltage and DC-link voltage underwent a rapid collapse, resulting in loss of sustainable motor operation.

This behaviour was investigated separately to identify whether the observed limitation originated from the motor-control system, insufficient DC-link energy storage, or the finite power capability and nonlinear characteristics of the PV source.

The investigation was carried out progressively by examining the system response near the critical operating speed, varying the DC-link capacitance, and replacing the motor-drive load with an equivalent resistive load to isolate the source-side dynamics. This allowed the collapse mechanism to be distinguished from the transient behaviour of the induction motor and its control system.

The analysis presented in this chapter demonstrates that the observed operating boundary is fundamentally associated with the power balance of the standalone system and the interaction between the boost converter and the nonlinear PV source.

6.1 Identification of the Maximum Sustainable Operating Speed

During evaluation of the complete standalone PV-fed induction motor pumping system, a distinct operating boundary was observed as the motor speed reference was increased. The system remained capable of sustaining stable operation up to approximately

$$\omega_m^* = 128 \text{ rad/s}$$

At this operating point, the motor was able to maintain the commanded speed while the PV source and boost converter continued to supply the required power to the drive.

When the speed reference was increased slightly further to

$$\omega_m^* = 129 \text{ rad/s}$$

the system was no longer able to sustain the operating point. Instead of settling at the commanded speed, the PV-side voltage and consequently the DC-link voltage underwent a rapid reduction, resulting in collapse of the power supplied to the motor drive.

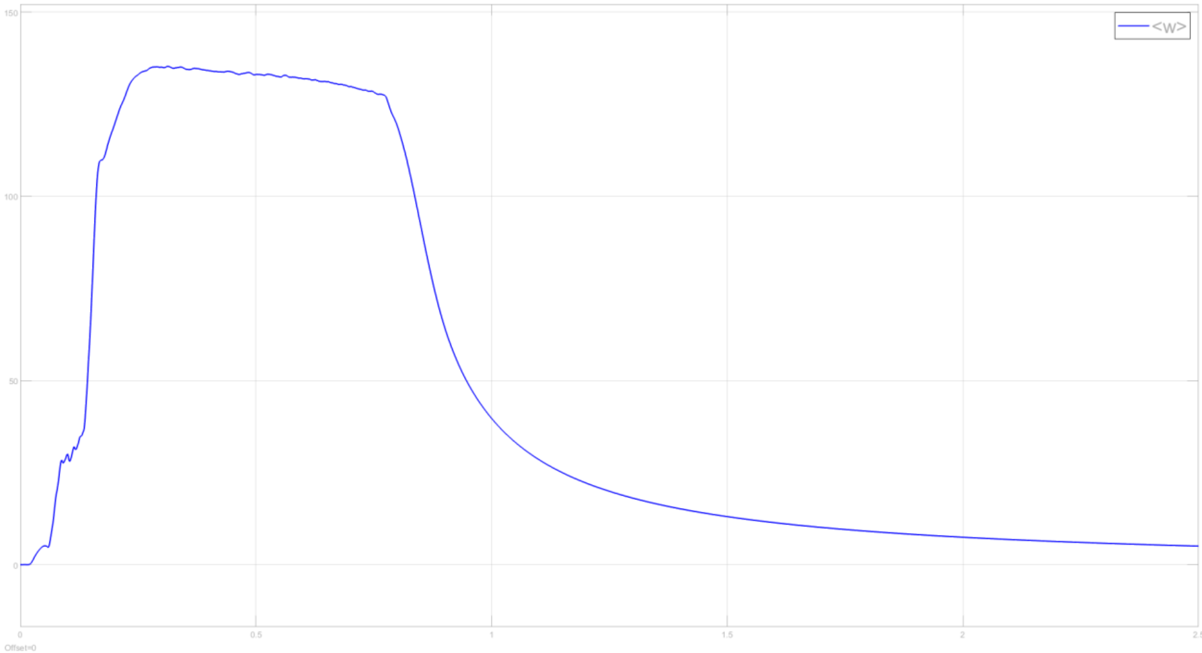

Figure 6.1: Motor Speed Response at a Reference Speed of 129 rad/s

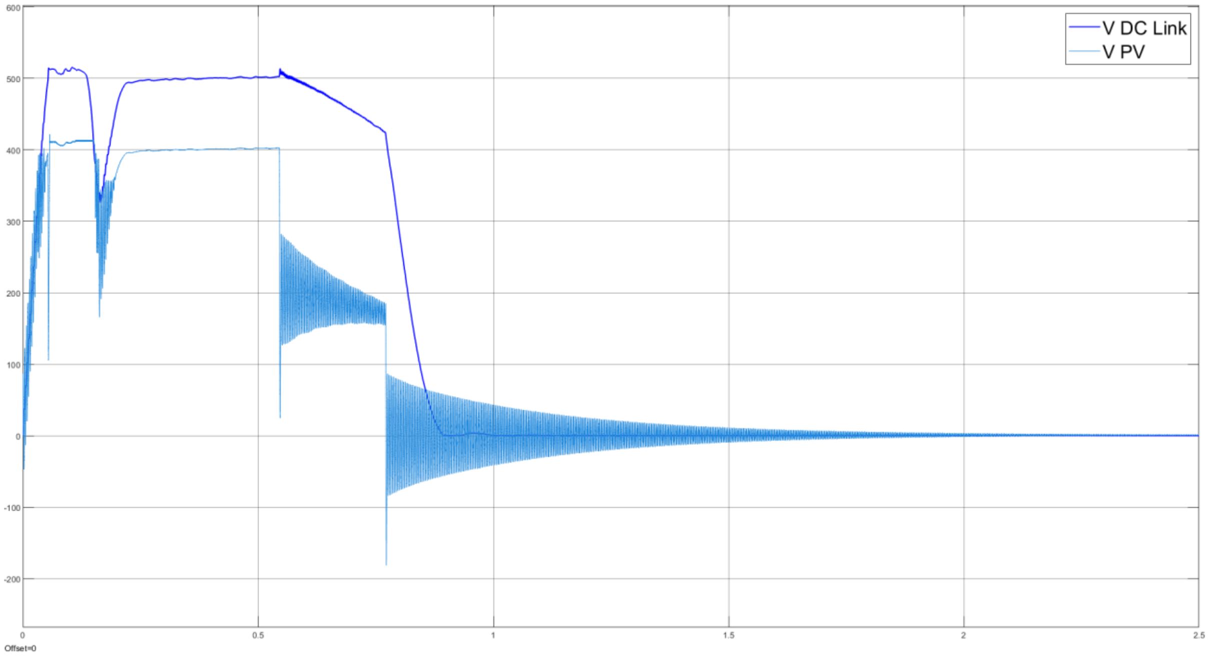

Figure 6.2: PV and DC-Link Voltage Collapse at 129 rad/s

The sharp transition between the two operating points is associated with the nonlinear relationship between motor speed and centrifugal pump power demand. For a centrifugal pump

$$T_L = K\omega_m^2$$

and therefore

$$P_L = T_L\omega_m$$

giving

$$P_L = K\omega_m^3$$

Thus, although the increase from 128 rad/s to 129 rad/s appears small, the required pump power increases according to the cube of motor speed. Near the maximum power capability of the standalone PV source, even a relatively small increase in speed can therefore move the system from a sustainable operating point to one at which the demanded power can no longer be continuously supplied.

To determine whether the observed collapse was primarily caused by insufficient energy storage in the DC link, the simulation was repeated with an increased DC-link capacitance. Although the larger capacitance affected the transient response and provided additional short-term energy buffering, it did not eliminate the collapse. This indicated that the phenomenon was not fundamentally caused by insufficient DC-link capacitance.

The motor drive was also replaced by an equivalent resistive load to isolate the PV source and boost-converter dynamics from the induction motor and IFOC system. Similar source-side collapse behaviour was observed when the loading exceeded the sustainable operating region. This confirmed that the operating limit was not primarily caused by the motor-control system but was associated with the interaction between the power demand, boost converter, and finite power capability of the PV source.

The physical mechanism responsible for this source-side collapse is examined in the following section.

6.2 Physical Mechanism of PV Source-Side Collapse

The collapse observed beyond the sustainable operating point is governed by the nonlinear electrical characteristic of the PV array. For a fixed irradiance and temperature, the PV array has a maximum power that can be extracted at a particular operating voltage. Increasing the electrical loading beyond this point does not result in a further increase in the generated power.

When the demanded power exceeds the available PV power, the deficit causes the DC-link voltage to decrease. The DC-link controller responds by adjusting the boost-converter duty ratio in an attempt to restore the required voltage. This action increases the effective loading imposed on the PV array and shifts its operating point toward the lower-voltage side of the P-V characteristic.

Beyond the maximum-power point, further reduction in PV voltage results in a reduction in generated power. Consequently, the corrective action of the converter can no longer restore the required power balance and instead drives the PV operating point further into the low-voltage region.

The collapse mechanism can therefore be summarized as

$$V_{DC} \downarrow \rightarrow D \uparrow \rightarrow V_{PV} \downarrow \rightarrow P_{PV} \downarrow \rightarrow V_{DC} \downarrow$$

Thus, once the source-load equilibrium is lost, the boost converter cannot compensate for the power deficit through duty-ratio control alone. Instead, the interaction between the DC-link controller and the nonlinear PV characteristic produces a regenerative reduction in the source voltage and available power, leading to the observed rapid voltage collapse.

This behaviour establishes the practical source-side operating boundary of the developed standalone system under a given set of environmental conditions.

7. Conclusion and Future Scope

The developed project presents the modelling and control of a standalone solar PV-fed induction motor drive for a centrifugal water-pumping application. The complete system integrates a PV array, DC–DC boost converter, DC-link capacitor, three-phase voltage-source inverter, induction motor, and centrifugal pump load.

The PV array and power-converter components were designed according to the rated requirements of the motor-drive system. Indirect Field-Oriented Control was implemented to achieve controlled operation of the induction motor by independently establishing the flux-producing and torque-producing current references. A speed controller designed using Bode-plot loop shaping was used to generate the required electromagnetic torque reference, from which the corresponding motor-control quantities and three-phase voltage references were obtained.

On the source side, separate small-signal models of the boost converter were used to design the PV-voltage and DC-link voltage controllers. A supervisory control strategy was developed to coordinate the two operating objectives. The minimum DC-link voltage required by the motor drive was calculated dynamically from the instantaneous three-phase voltage references. Based on the available DC-link voltage margin, the supervisory controller selected either DC-link voltage support or maximum-power-point operation. Hysteresis and mode-dependent controller activation were incorporated to improve the transition between the two operating modes.

Simulation results demonstrated successful operation of the integrated system under feasible operating conditions and its response to variations in irradiance, temperature, and motor speed reference. The motor-drive system was able to track the commanded operating speed while the PV-side converter supplied the required power and maintained the voltage necessary for inverter operation.

The investigation of the system operating limit further demonstrated the importance of source-load power balance in a standalone PV-fed drive. A distinct maximum sustainable operating region was observed, beyond which the interaction between the boost converter and the nonlinear PV-source characteristic resulted in source-side voltage collapse. Additional investigations indicated that this limitation was associated with the finite power capability of the PV source rather than insufficient DC-link capacitance or the induction motor control system.

The present work can be extended by incorporating explicit DC-link overvoltage management and PV power-curtailment control for conditions in which the available solar power exceeds the instantaneous motor-load requirement. Energy-storage integration may also be considered to absorb surplus PV power and provide additional support during temporary source-power deficits. Further development may include adaptive maximum-power-point tracking, experimental validation of the proposed control architecture on a hardware prototype, and implementation of the control algorithms on a real-time embedded control platform.

Overall, the project demonstrates an integrated control approach for a standalone PV-fed induction motor pumping system while also identifying the practical operating limitations imposed by direct coupling between a nonlinear renewable-energy source and a dynamic motor load.